2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N) | 978-1-6654-3811-7/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICAC3N53548.2021.9725457

# Web Browser Forensics for Retrieving Searched Keywords on the Internet

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***Abstract*—Nowadays, most of the reported cybercrimes use the Internet. It is seen that Web Browsers leave forensically sound footprints of Internet browsing activities on the computer's storage media. The traces once retrieved stand as crucial hints that help the investigator to reconstruct various events that the suspect has done. However, the usage of private or portable web browsers prevents the creation of those files on the Suspect's Computer. This makes the job of investigators more challenging. But, the remnants of browsing activities done using any web browser is recorded in the Physical Memory of the Computer. Hence, analyzing the Physical Memory Dump of the suspect's machine, collected as part of Live Forensics, may provide lots of forensically sound evidence related to the Internet-related activities in a cybercrime investigation. This information provides highly crucial information which can render hints in the investigation process. One of the most important features of any Cyber Forensics Analysis Tool is keyword searching. But identifying and finalizing the keywords to be searched during the analysis process in a particular Cyber Crime is always challenging for the investigators. The task of retrieving searched keywords is crucial as it directly provides the criminal background of the Suspect if any. This information enables cyber forensics investigators to conduct a focused approach while analyzing the evidence. This paper mainly presents a novel methodology for the reconstruction of searched keywords from the Physical Memory Dump collected from the Suspect's Windows 10 Computers. A methodology for retrieving keywords from the browser files generated by Web Browsers in the storage media has also been described. The keywords retrieved in this way helps the Investigators to identify the associated highly crucial information during the deep-dive analysis of the storage media in Offline Forensic Analysis.**

***Keywords- Browser Forensics, Cyber Forensics, Live Forensic, Memory Forensics***

## I. INTRODUCTION

Cyber Forensics mainly involves processes to acquire and analyze forensically sound digital evidence to present before the court of law. It usually starts with acquiring digital data from the storage devices at the scene of crime and analysing those data at the analysis lab to retrieve crucial evidence. This area involves a lot of challenges concerning the advancements in technologies. Also, the information required by the investigators differs based on the nature of the crime. And the evidence available on a computer varies depending on the type and version of the operating system in it. As new versions of operating systems launch, the nature of available evidence also varies. Usually, a forensics investigator chooses Live or Offline Forensics based on the state of the systems at the scene of crime. If the computer is in the 'on' state, the analyst starts with a Live Forensics approach. On the other hand, for computers that are already switched off, the only option is to go ahead with offline traditional forensics. In the case of offline forensics, the analyst first takes a copy of the bit-stream content of storage media and analyze it later to retrieve forensically sound evidence.

## II. LIVE AND OFFLINE FORENSICS

### *A. Live Forensics*

Live Forensics acquires and analyzes volatile data present in a running computer. This provides forensically sound evidence which will be lost forever once the system is switched off. This is a newly recommended forensic procedure if the Suspect's machine is in the 'on' mode at the scene of crime. On the other hand, adopting a traditional forensic approach in a running system results in losing all the volatile information forever. In Live Forensics, the content of RAM is collected to a file, which will be analyzed later to reconstruct the evidence related to ongoing activities in the Suspect's Computer. The information collected in this way includes the details of running processes and its associated details like files opened, network connections, dynamic link libraries used, command-line information etc. This gives a clear picture of ongoing activities happening on a Computer. It also reveals the system-level information and details of events recently happened in the system. It may also provide encrypted data in raw form and encryption keys, which may be useful in decrypting the encrypted data present in the storage media. Moreover, modern-day crimes are either organized crimes or malware initiated attacks. In these cases, usually, there won't be any information related to the crime inside the hard disks. However, Physical Memory may hold the footprints of the activities related to the crime. Thus Live Forensics is highly recommended if the system is in the 'on' state at the scene of crime.

### *B. Offline Forensics*

Traditional Offline Forensics acquires the bit-stream copy of the storage media and analyzes it to extract evidence. The analysis is mainly done to reconstruct files and folders in the Suspect's machine through a file system level analysis. Then, the content of these reconstructed files will be analyzed to retrieve crucial evidence. This results in recovering deleted and overwritten files, retrieving data from the unallocated parts and slack areas of the disk. Different types of analysis techniques applied to the recovered files reveal more useful evidence. The types of analysis involve Keyword Searching, File Searching, Timeline Analysis, Bookmarking, Email Analysis,

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Operating System Artifacts Analysis, Signature Mismatch Analysis etc.

### *C. Browser Forensics*

Usually, some files are generated in the local system when a user browses the Internet [1] using web browsers. These files contain details of visited websites, cached information, cookies data, bookmarked websites etc. Whenever a user browses some websites, the details of that visit will be recorded in these locally generated browser files [2]. These files are extremely useful for cyber forensics investigations as they may contain useful evidence related to the reported crime. However, if the suspect uses some private or portable browsers, there won't be any data generated in the browser files [3]. Also, in cases of planned crimes, committed by criminals having enough expertise in the cyber forensics area, the browser files and associated information in the files will be erased in advance using anti-forensics techniques. This makes the job of a cyber forensics investigator more challenging. Even though there are no traces available in the browser files, the traces of the browsed data may be recorded in the physical memory of the system. Another common observation made in most of the crimes reported is the anti-forensics techniques adopted to purposefully delete the evidence that may be available in the Suspect's computer. The anti-forensics techniques adopted may include the usage of file shredding tools to wipe the entire content of the storage media or any other tools to alter the evidence that may mislead the investigation. However, conducting Live Forensics enables the extraction of crucial browser-related data from the physical memory which may not be available anywhere on the hard disk.

Social Media based crimes are very common nowadays. The suspicious keywords used by the suspect in the social media may provide crucial hints to directly link him to the reported cyber crime. Those details can also be collected from the memory dump in case the browser files are not created in the Suspect's machine. In addition to the usage of private and portable browsers, another common method used to hide the browsed content is to disable the settings to stop the creation of browser files. The following section explains a novel methodology for retrieving keywords searched by the suspect in various web browsers from RAM content. This algorithm works fine for analyzing the physical memory dump collected from all Windows Computers. Subsection A describes the acquisition of the memory content to a file. The novel methodology developed for identifying and extracting searched keywords from the Windows memory dump is described in subsection B.

## III. Memory Forensics

### *A. Acquisition of Physical Memory Content*

The first step in Memory Forensics is to acquire the bit-stream copy of the memory content to a file[4]. During the analysis process, this file is to be analyzed on the Investigators' machine, to retrieve forensically important evidence. There are software or hardware-based approaches available to collect the physical memory content. However, collecting the memory dump using software is widely accepted. Figure 1 shows acquiring the content of physical memory using the software tool DumpIt [1]. The following section depicts the details of recovering searched keywords on the Internet from the collected memory dump.

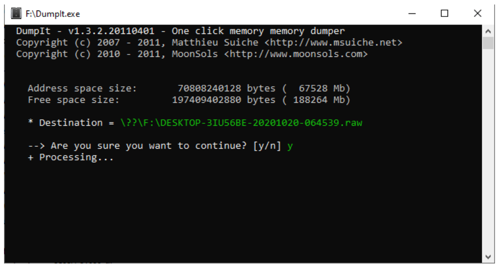


Fig. 1. Collecting Physical Memory dump using DumpIt.

### *B. Analysis of the Memory Dump File*

Physical Memory Analysis is an important step in Memory Forensics where the memory dump collected from the Suspect's machine is analyzed by reading it block by block for extracting the evidence. The analysis of the memory dump file provides information regarding the user's Internet-related activities [5] also. This type of browser forensics may provide more recent information related to the Suspect's activities compared to the information that can be collected by analyzing the browser files created in the local system. Nowadays, a memory dump file is of gigabytes (GB) in size. Usually, a block of searched terms in plain text format is found in the memory dump. But, finding keywords by manual observation of the file is a tedious job because of the huge size of the memory dump. Research conducted to retrieve the list of searched keywords automatically from the memory dump reveals that the physical memory of recent Windows OS Computers stores these keywords in two different formats. In the novel methodology described in this paper, the keywords are identified based on the two formats found in the memory. In this method, all the keywords can be retrieved by searching for the fixed patterns. In the first format, the keywords are found after some specific headers or hexadecimal patterns. Here, two types of patterns may be available in the memory, one is a 6-byte pattern and the other is a 7-byte pattern. The first pattern is 'X 04 01 Y 02 02' and the other is 'X 05 01 81 Y 02 02'. The following section depicts the format of these patterns and algorithms to decode them automatically. The second format in which search terms are available in the memory dump file is by storing the terms after some specific keywords.

In Figure 2, the 6-byte patterns found are '26 04 01 4B 02 02', '18 04 01 2F 02 02', '28 04 01 4F 02 02', '0D 04 01 19 02 02', '0F 04 01 1D 02 02' and '29 04 01 51 02 02'. In these hexadecimal patterns, all the bytes are the same except the first and fourth bytes. Keywords are found after this pattern. From our experiments, it is observed that the first byte in the pattern is the number of bytes to be read to get the searched term. For example, in the pattern '0F 04 01 1D 02 02', the number of bytes to be read is 0F, which is 15. While reading the 15 bytes starting from the beginning of this pattern, the search keywords can be identified. In Figure 2, the keyword obtained, 'facebook' is highlighted. The keyword is followed by a byte '02' which marks the end of the string. Similar methodology can be applied in retrieving the searched terms associated with other 6-byte patterns also. Another important point observed in our research is that the fourth byte in the 6-byte pattern is one less than double the first byte. In the selected pattern, '0F 04 01 1D

02 02', the first byte is ‘0F’ and thus the fourth byte should be, (2*0F)-1, which is ‘1D’. The observation is true for all the similar patterns.

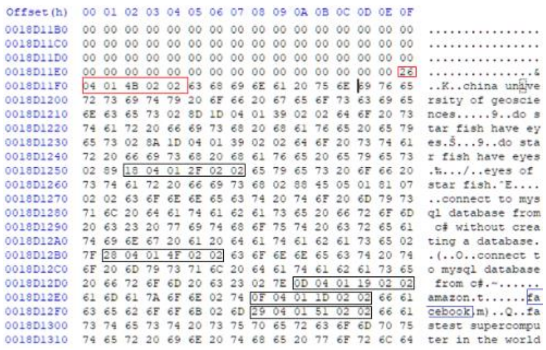

Fig. 2. Searched Keywords 6-byte Pattern ‘X 04 01 Y 02 02’

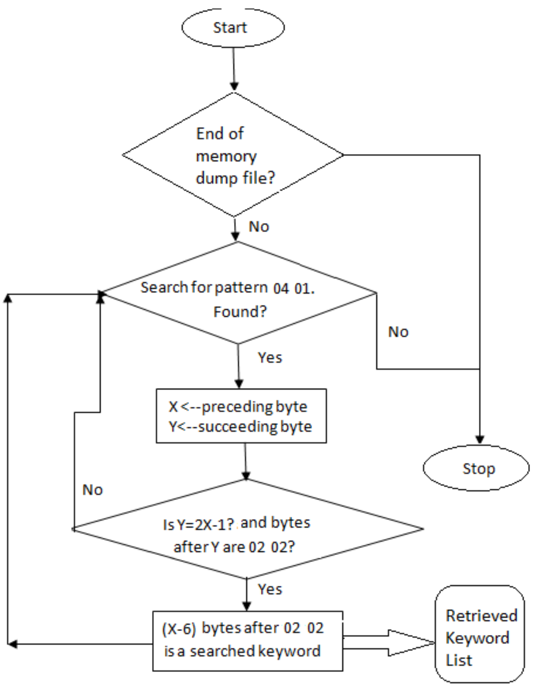


Fig. 3. Flowchart for extracting keywords from memory dump file in case of 6-byte pattern ‘X 04 01 Y 02 02’

So, the format of this pattern can be generalized as shown below.

X 04 01 Y 02 02

Here, X is the first byte and Y is the fourth byte which must be of value 2X-1. Here, X is the number of bytes to be read from the offset of the pattern as mentioned above. The searched keywords start after the pattern and end with hex value 02. A flowchart for retrieving keywords by this method is given in figure 3. And the algorithm for finding the search keywords is given below.

**Step 1:** Search for pattern 04 01 in the memory dump file.

**Step 2:** Let X be the preceding byte and Y be the succeeding byte. Check whether Y=2*X-1 and 5th and 6th bytes are 02 02.

**Step 3:** If No, go to Step 1.

**Step 4:** If Yes, read X number of bytes from the starting of the pattern. Here (X-6) bytes after the pattern gives the searched keywords.

The second pattern observed is a 7-byte pattern as shown in Figure 4. The patterns highlighted in the figure are ‘69 05 01 81 4F 02 02' and '45 05 01 81 07 02 02'. Here, all the bytes are the same except the first and fifth bytes. From our experiments, it is observed that the first and fifth byte holds a relation in all the similar patterns. And similar to the 6-byte pattern, the first byte is the number of bytes to be read to get the searched keyword. For example, consider the pattern '45 05 01 81 07 02 02' in Figure 4. The first byte is ‘45’, i.e. ‘45’ is the number of bytes to be read from the beginning of the pattern. Similar to the 6-byte pattern method, the end of the searched keyword is marked with ‘02’. Thus, after the pattern, we get the string "connect to the MySQL database from C# without creating a database" with 02 as the end of the string marker as shown in Figure 4. This 7-byte pattern can be generalized as shown below.

X 05 01 81 Y 02 02

Here, X is the first byte and Y is the fifth byte. Our research shows that Y holds a value equal to 2X-83. For example, in the pattern '45 05 01 81 07 02 02' shown in Figure 4, the fifth byte of the pattern is 07 which can be interpreted as Y=2X-83 which is 2X-(81(4$^{th}$ byte) + 02(6$^{th}$ byte)).

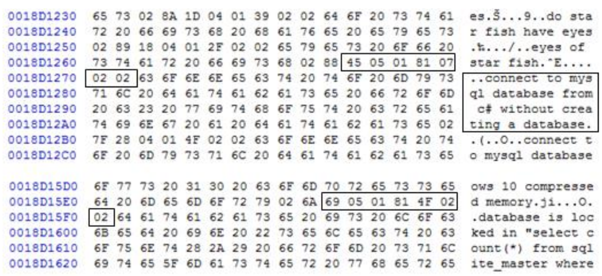

Fig. 4. Searched Keyword- 7-byte pattern ‘X 05 01 81 Y 02 02’

The following algorithm describes how the searched keywords can be retrieved from the memory dump using the 7-byte pattern.

**Step 1:** Search for pattern 05 01 81 in the memory dump file.

**Step 2:** Let X be the preceding byte and Y be the succeeding byte. Check whether X*2-2-81=Y and 6th and 7th bytes are 02 02.

**Step 3:** If No, go to Step 1.

**Step 4:** If Yes, read X number of bytes starting from the beginning of the pattern. Here (X-7) bytes after the pattern gives the searched keywords.

A flowchart for this method is given in figure 5.

So the hexadecimal patterns identified in the new methodology can be summarized as two patterns

‘X 04 01 Y 02 02’ where Y=2X-1 and

‘X 05 01 81 Y 02 02’ where Y=2X-83

It can be mentioned as

'X 04 01 2X-1 02 02'

'X 05 01 81 2X-83 02 02'

In both of the cases, the first byte specifies the number of bytes to be read to extract the keywords.

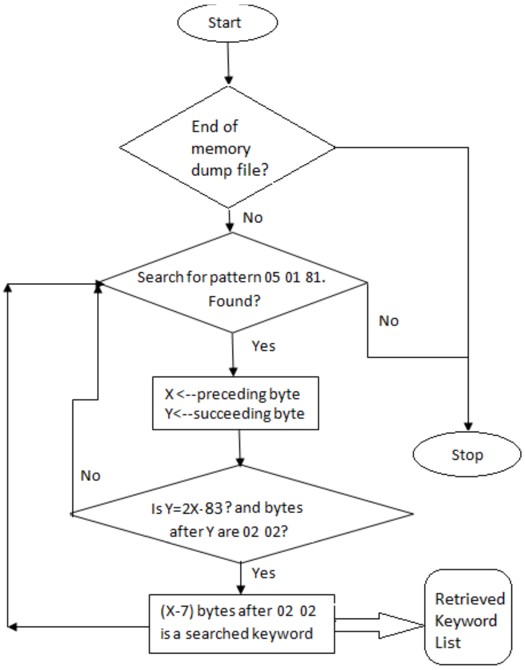


Fig. 5. Flowchart for extracting keywords from the memory dump file in case of 7-byte pattern 'X 05 01 81 Y 02 02'

The second format in which searched keywords are available in the memory dump file is by storing the keywords after some specific keywords. In this case, the search queries can be retrieved from memory by searching for these specific keywords. These specific keywords after which search terms are stored in Windows Computers are 'search_query=' and 'search?q='. The words in the search query searched by the Suspect are separated by special character '+'. In the case of 'search_query=', the query ends with hex value 00. Figure 6 shows the keyword found using this method. And the end of the string is marked by '&' in case of 'search?q='. A keyword found in this way is shown in Figure 7. Thus the list of keywords searched by the Suspect can be retrieved in this way also. All these directly support cybercrime investigation.

The searched terms retrieved from memory forensics using the above methodologies can be used during offline forensics to identify associated details. This is usually done by conducting a keyword search on the files and folders recovered from the storage media. By an in-depth analysis of these results, a clear picture and motive behind the committed crime can be identified.

Identifying URLs browsed by the Suspect is also important in a cyber crime investigation. This can be done by searching the memory dump for 'http://' or 'https://', the URLs browsed by the Suspect can be retrieved. Figure 8 shows such hits in the memory dump file. But one observation found in our research is that, in the case of a few sites, it is stored in an encoded form. The special characters like ':' and '/' are replaced by its ASCII value. The ASCII value of ':' is 72 and the corresponding hex value is 3A. In the case of '/', the hex value is 2F. So the URL is saved into the memory as "https%3A%2F%2F" by replacing 'https://' A URL obtained in this format is displayed in Figure 9. Another format in which URL is found is using a double encoded form where the '%' in the above-mentioned URL is again encoded in the same way. Here '%' is replaced by '%25'. Figure 10 shows such a hit in the memory dump. Here 'https://' is replaced with 'https%253A%252F%252F'.

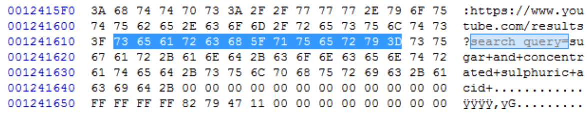


Fig. 6. Keywords found in the memory dump file in the 'search_query=...+..+..+......' format

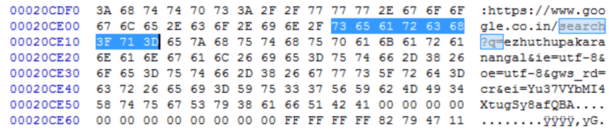


Fig. 7. Keywords found in memory dump file in the 'search?q=...+...+...&' format.

## IV. BROWSER FILES IN OFFLINE FORENSICS

The Searched Keywords can also be retrieved during Offline Forensics. Here, browser files generated locally in the hard disks are to be analysed[6]. All browsers store browsing information in files in the 'Users' folder in the operating system drive on a Windows Computer. There are unique folders generated for each Web Browser. Google Chrome, Mozilla Firefox and Opera browsers store information about searched keywords in these browser files in SQLite database formats. These SQLite files are created in the \\Users\UserName\AppData\Browser_Path. The Browser Path for different browsers are given in Table I.

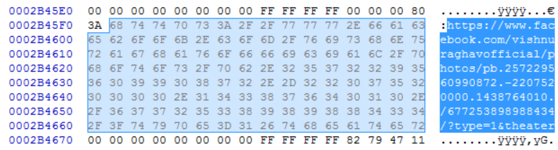


Fig. 8. URL in Windows 10 memory dump starting with 'https://'

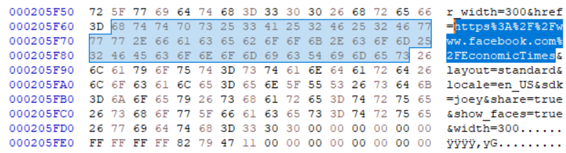


Fig. 9. URL in https:// is encoded as "https%3A%2F%2F"

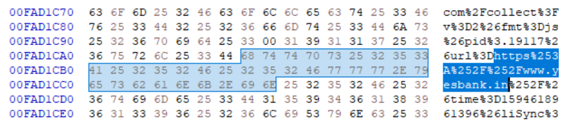


Fig. 10. URL in https:// is encoded as "https%253A%252F%252F"

The Chrome browser stores the details of searched keywords in the SQLite database named 'history.sqlite' found in the path \\Local\Google\Chrome\User Data\Default\. The keywords are stored inside the table 'keyword_search_terms' in this database. The table contains information like search terms and URLs. The Firefox browser stores the details of searched keywords in SQLite database named 'places.sqlite' found in the path '\\Roaming\Mozilla\Firefox\Profiles\ <random text>.default\'. The keywords are stored inside the table 'history', in this database. The table holds searched terms and the associated URL. The Opera browser stores the details of searched keywords in the SQLite database named 'history.sqlite' found in the path '\\Roaming\Opera Software\Opera Stable\'. The keywords are stored inside the table 'keyword searched terms' in this database. The table contains searched keywords and the corresponding URL. Table I shows the details of browser files holding the searched keywords in different browsers.

TABLE I. DETAILS OF SEARCHED TERMS IN BROWSER FILES

| Web Browsers | Browser Path | File Name | Details Available |
|---|---|---|---|
| Google Chrome | \\Local\Google\Chrome\User Data\Default\ | history.sqlite | Searched Terms, Visited URL |
| Mozilla Firefox | \\Roaming\Mozilla\Firefox\Profiles \ <random text>.default\ | places.sqlite | Searched keywords, Visited URL |
| Opera | \\Roaming\Opera Software\Opera Stable\ | history.sqlite | Searched Terms, Visited URL |

## V. COMBINING KEYWORDS OBTAINED FOR KEYWORD SEARCHING IN OFFLINE FORENSICS

As mentioned in the previous section, one of the challenges faced by Cyber Forensics investigators is in deciding the keywords to be searched while analyzing the storage media content retrieved as part of the Offline Forensics. This is because whenever an in-depth analysis is required, the list of keywords to be searched is to be decided by the investigators. So the keywords obtained through algorithms mentioned in sections III and IV in the paper can be merged with the case related keywords to create the list of keywords to be searched in the cyber forensics analysis. Figure 11 gives a module diagram describing the processes to identify the crime-related files and folders from offline forensics by doing keyword searching. The searching is conducted using the keywords recovered with the methodologies explained in this paper.

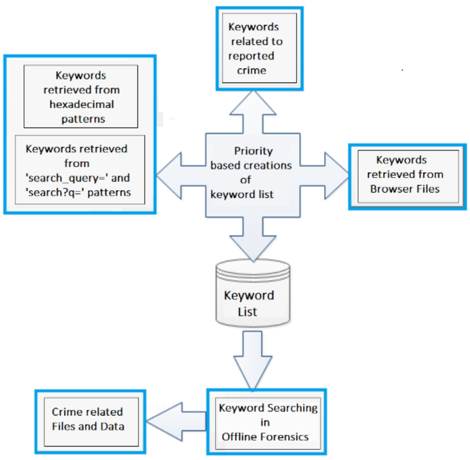


Fig. 11. Module diagram showing the extraction of crime-related files in Offline Forensics based on the keywords identified

## VI. CHALLENGES AND FUTURE WORK

The main challenge while adopting this methodology is that, in a considerable number of cases, the computer will be in the shutdown state. Then the physical memory dump cannot be collected and the type of aanalysis as described in the paper is not possible. A solution that can be proposed in those cases is to decode the hibernation file available in the operating system drive to get a memory dump file. A similar analysis can be possible once the memory dump is reconstructed from the hibernation files. But the limitation here is that the content of the created memory dump will be the one at the time when the system last hibernates. So there is no guarantee that it may contain the recent browser activities of the suspect.

## VII. CONCLUSION

Cyber Forensics poses many challenges to the Investigators as new technologies evolve. Specifying the keywords to be searched in the analysis tool is often challenging to the Investigators. The list of keywords retrieved from the memory dump using the novel methodology described in this paper enables the investigator to quickly collect crime-related keywords searched by the Suspect on the Internet. The investigator can then use these keywords to find crime related files in the storage media having the same keywords. This directly helps in predicting whether the suspect is intentionally involved in the reported internet related cyber crime or not. Hence, the described methodology provides crucial hints to the investigator regarding the browsing habits of the suspect.

## REFERENCES

[1] Abner Mendoza, Avinash Kumar, David Midcap, Hyuk Cho, and Cihan Varol, "BrowStEx: A tool to aggregate browser storage artifacts for forensic analysis," Digital investigation, September 2015.

[2] David Mugisha, "WEB BROWSER FORENSICS : Evidence Collection and Analysis for Most Popular Web Browsers usage in Windows 10," Thesis in International Journal of Cyber Criminology.

[3] A. Reed1, M. Scanlon2, N-A. Le-Khac, "Private Web Browser Forensics: A Case Study of the Epic Privacy Browser," Journal of Information Warfare, Vol-17, January 2018.

[4] Dija S, Indu V, Sajeena A, Vidhya J A, "A Framework for Browser Forensics in Live Windows Systems," 2017 IEEE International Conference on Computational Intelligence and Computing Research.

[5] Neethu Joseph, Sherina Sunny, Dija S, Thomas K L, "Volatile Internet Evidence Extraction from Windows Systems," 2014 IEEE International Conference on Computational Intelligence and Computing Research.

[6] Mayur Rajendra Jadhav, Dr. Bandu Baburao Meshram, "Web Browser Forensics for Detecting User Activities," July 2018.

[7] Sweta Mahaju, Travis Atkison, "Evaluation of Firefox Browser Forensics Tools," in Proceedings of ACM SE '17, Kennesaw, GA, USA, April 13-15, 2017.